

5. GFR Estimation — Formulas

Cockcroft-Gault:

$$\text{CrCl} = (140 - \text{age}) \times \text{weight (kg)} / (72 \times \text{PCr}) \times 0.85 \text{ if female}$$

MDRD:

$$\text{eGFR} = 186 \times \text{PCr}^{-1.154} \times \text{age}^{-0.203} \times 0.742 \text{ if female}$$

Which to use:

- MDRD more accurate at GFR <60 mL/min/1.73m²
- Cockcroft-Gault still used for drug dosing

Critical rule — applies to both: These formulas assume **steady state** (stable creatinine). In AKI with rising creatinine day-to-day, **neither formula is valid**. Do not use for drug dosing in active AKI.

Low muscle mass trap: Elderly frail patient, bedridden, on steroids → creatinine stays falsely low → eGFR overestimates true GFR → true GFR may be <30 even with "normal" creatinine of 0.8.

1. Cockcroft-Gault equation

WHAT YOU SEE

$\text{CrCl} = (140 - \text{age}) \times \text{weight}(\text{kg}) / (72 \times \text{PCr}) \times 0.85$ if female

WHAT IT ENCODES

Cockcroft-Gault estimates creatinine CLEARANCE in mL/min (absolute, NOT indexed to 1.73m^2) from age, weight, sex, and serum creatinine. It systematically OVERESTIMATES true GFR because creatinine is also tubularly secreted. Despite being older and less accurate, it remains the formula behind most FDA drug renal-dosing labels.

BOARD TRAP

Use eGFR (MDRD/CKD-EPI) for renal drug dosing. WRONG for many drugs — most renal dosing thresholds were derived from Cockcroft-Gault creatinine clearance (mL/min); using a BSA-indexed eGFR can misclassify dosing, especially at extremes of body size.

2. $\times 0.85$ female factor

WHAT YOU SEE

$\times 0.85$ if female

WHAT IT ENCODES

Women have lower muscle mass and thus lower creatinine generation, so C-G multiplies by 0.85 (a 15% reduction) for females. The analogous sex factor in MDRD is $\times 0.742$.

BOARD TRAP

Forgetting the 0.85 female factor overestimates a woman's clearance by $\sim 18\%$, potentially leading to renal-cleared drug overdosing.

3. Weight in C-G

WHAT YOU SEE

$\times \text{weight}(\text{kg})$

WHAT IT ENCODES

Cockcroft-Gault uniquely uses absolute body WEIGHT, which is why it gives an un-indexed mL/min. In obesity or edema, actual body weight inflates the estimate; use ideal body weight (or adjusted body weight if obese) for dosing. In underweight patients use actual weight.

BOARD TRAP

Plug actual body weight into C-G for an obese patient. WRONG — fat and edema fluid don't generate creatinine, so actual weight overestimates clearance; use ideal/adjusted body weight.

4. MDRD equation

WHAT YOU SEE

$\text{eGFR} = 186 \times \text{PCr}^{-1.154} \times \text{age}^{-0.203} \times 0.742$ if female

WHAT IT ENCODES

MDRD reports eGFR INDEXED to 1.73m^2 BSA (mL/min/ 1.73m^2) using creatinine, age, and sex (the race coefficient has been removed in current practice). It is validated for CKD staging and more accurate than C-G for GFR <60 .

BOARD TRAP

MDRD clearance and C-G clearance are interchangeable. WRONG — MDRD is BSA-indexed (per 1.73m^2) while C-G is absolute mL/min; converting requires de-indexing by the patient's actual BSA before using for drug dosing.

5. MDRD more accurate <60

WHAT YOU SEE

MDRD more accurate for GFR <60 mL/min/ 1.73m^2

WHAT IT ENCODES

MDRD was derived in a CKD population, so it is reliable for GFR <60 but UNDERESTIMATES (and labs won't report exact values for) GFR above 60. CKD-EPI is the modern preferred equation because it is accurate across the full range, especially at near-normal GFR, and reduces false CKD labeling.

BOARD TRAP

An MDRD eGFR of 'roughly 75' should be acted on as a precise value. WRONG — MDRD is unreliable above 60 and underreports near-normal GFR; use CKD-EPI when GFR is near normal, and don't call a healthy patient CKD on an inflated-downward MDRD.

6. C-G for drug dosing

WHAT YOU SEE

Cockcroft-Gault still used for drug dosing

WHAT IT ENCODES

Renal dose adjustments (e.g., DOACs like dabigatran/rivaroxaban, aminoglycosides, vancomycin, enoxaparin, gabapentin) were largely studied using Cockcroft-Gault CrCl, so guidelines and package inserts dose by C-G mL/min. CKD STAGING uses eGFR (CKD-EPI/MDRD).

BOARD TRAP

Board favorite split: DRUG DOSING $\rightarrow$ Cockcroft-Gault (CrCl, mL/min); CKD STAGING $\rightarrow$ eGFR (CKD-EPI/MDRD, mL/min/ 1.73m^2). Mixing them up is the classic distractor.

7. Critical rule — steady state

WHAT YOU SEE

Critical rule — both formulas assume steady-state (stable creatinine)

WHAT IT ENCODES

Every creatinine-based estimating equation assumes the creatinine is at STEADY STATE (stable day to day). The math is invalid the moment creatinine is rising or falling, because the equation can't account for the changing balance between production and clearance.

BOARD TRAP

Calculate an eGFR off a single creatinine in a patient whose creatinine jumped overnight. WRONG — non-steady-state creatinine makes eGFR/CrCl meaningless; you cannot estimate GFR during active AKI.

8. AKI invalidates formulas

WHAT YOU SEE

In AKI with rising creatinine day-to-day, neither formula is valid. Do not use for drug dosing in active AKI.

WHAT IT ENCODES

In AKI, true GFR can fall to near zero while serum creatinine is still 'normal' because creatinine accumulates with a 24–72h lag. Estimating equations therefore overstate GFR early in AKI; dose renally-cleared drugs conservatively (assume low GFR) and trend the creatinine instead.

BOARD TRAP

A normal creatinine on the first day of AKI means GFR is fine and full drug doses are safe. WRONG — creatinine lags GFR by days; true GFR may already be near zero, so estimating equations and standard dosing are unsafe in active AKI.

9. Low muscle mass trap

WHAT YOU SEE

Low muscle mass trap

WHAT IT ENCODES

Creatinine is generated by skeletal muscle, so low muscle mass produces a low serum creatinine independent of kidney function — making creatinine-based eGFR falsely HIGH. Affected groups: elderly, frail/bedridden, malnourished, cirrhotic, amputees, and patients with paralysis or on corticosteroids.

BOARD TRAP

A creatinine of 0.8 in a frail elderly woman means normal kidney function. WRONG — her low muscle mass keeps creatinine low; true GFR may be <30. Creatinine-based eGFR is falsely reassuring in low-muscle states.

10. Elderly frail overestimate

WHAT YOU SEE

Elderly frail, bedridden, on steroids → creatinine stays falsely low → eGFR overestimates true GFR

WHAT IT ENCODES

In sarcopenic/cachectic patients, a 'normal' creatinine (e.g., 0.8) can coexist with a true GFR <30. This causes silent renal drug overdosing (e.g., DOACs, gabapentin) unless you adjust. Cystatin C or a measured/timed urine creatinine clearance gives a truer estimate here.

BOARD TRAP

Dose a renally-cleared drug at full dose in a cachectic elderly patient because the creatinine is normal. WRONG — true GFR may be severely reduced; confirm with cystatin C-based eGFR or a measured creatinine clearance before dosing.

11. Cystatin C alternative

WHAT YOU SEE

(implied alternative to creatinine)

WHAT IT ENCODES

Cystatin C is a low-molecular-weight protein produced by ALL nucleated cells at a constant rate, freely filtered and not secreted — so it is independent of muscle mass, age, and sex. It is the preferred GFR marker when creatinine is unreliable (cachexia, amputees, cirrhosis, extremes of muscle mass); a combined creatinine + cystatin C eGFR (CKD-EPI 2021) is the most accurate.

BOARD TRAP

Cystatin C is a flawless GFR marker. WRONG — it is falsely RAISED by corticosteroids, thyroid dysfunction (hyperthyroidism), high cell turnover, smoking, and inflammation/obesity, so it can overestimate the degree of renal impairment in those settings.
